Nitrogen System Review

Purchased or generated, nitrogen is set once at commissioning and then runs whether the line does or not. We measure the difference.

What it is

Inflow maps the nitrogen system from the bulk tank, cylinder manifold or generator to every point of use, builds a 12 month cost baseline, measures what flows while nothing is running, tags leaks, and ranks the fixes with dollars on each. Same method Inflow has used on compressed air for 18 years.

Where nitrogen is wasted

- **Laser assist, fab shops.** Assist pressure and purity set above what the cut or the edge needs. Booster and hose leaks at 300 psig.
- **Soldering atmosphere.** Reflow and selective solder atmospheres set to a conservative oxygen ppm at commissioning and running while the line idles. Idle time is often half the hours.
- **Tank blanketing and purge.** Blanket regulators drift up, conservation vents leak, purge points stay on between batches. Two or three vessels usually take most of the gas.
- **Packaging flush and LN2 dosing.** Modified atmosphere flush and LN2 dosing set for a residual oxygen target and never revisited. A flush cut with no change in residual oxygen is common.
- **Lab LN2 and gloveboxes.** Dewars boil off, gloveboxes purge around the clock, and no one owns the total across accounts.

What the review covers

- **System map.** Tank or generator to every point of use, with regulator setpoints, pipe sizes, and the purity each use actually requires.
- **Cost baseline.** 12 months of invoices, or generator power and purity records, at your contract rate.
- **Idle draw.** Tank level or generator run data across a shutdown separates leaks, purge and boil off from production use.
- **Leak survey.** Ultrasonic survey with each leak tagged, sized and valued at delivered price.
- **Point of use survey.** Flow and setpoint per use versus the process specification: assist pressure tables, solder oxygen ppm, blanket pressure design basis, residual oxygen tests.
- **Report.** System map, baseline, tagged leak list, ranked fixes with savings and budget costs.

Proof of method

Same method, first applied to argon at a Twin Cities medical device plant in 2026: 80 percent of purchases were idle draw from four small leaks, open quick disconnect whips and tank boil off. About $8,700 per year recoverable on a $4,700 review. Nitrogen systems show the same pattern at larger volumes.

Fee by scope

Review	Typical plant	Fee
Shielding and laser assist gas review	Fab shop, one or more lasers	$6,900 ($4,700 small shop)
Soldering atmosphere review	One or two lines	$3,900 ($5,900 three or more)
Blanketing and purge review	Multi tank	$6,900 ($4,700 single system)
Packaging gas review	One or two lines	$4,700 ($6,900 plant wide)

All inclusive, two site visits, report. Delivered nitrogen carries no utility rebate, so the fee pays back out of the gas bill. Generated nitrogen is different: the generator's compressor is a compressed air load and Xcel Energy study funding can apply. Inflow is an Xcel listed study provider and handles the application.

How it works

- **1. Observe.** Walk the system in production. List every use, its setpoint, and the specification it is supposed to meet.
- **2. Measure.** Invoices or generator data, an idle period reading, flow and oxygen checks at the points of use, and an ultrasonic pass on the lines.
- **3. Report.** Findings, tagged leaks, ranked fixes with dollars and budget costs, in a document your team, your gas supplier and your utility can all use.

Typical schedule: two site days plus one idle period reading. Draft report within two weeks of the second visit.

What you provide

- Escorted access to nitrogen lines and equipment
- 12 months of invoices, or generator power and purity records, and the contract rate
- Process specs: cut parameter tables, solder profiles, blanket design basis, residual oxygen specs
- Piping drawings if any, and a shutdown window for the idle reading

Credentials
Xcel Energy Energy Efficiency Partner Award winner, 12 consecutive years, 2015 to 2026
U.S. Department of Energy qualified AirMaster+, Pump and Fan System Specialist
Independent. Vendor neutral. We measure it, we do not sell it. 20+ years on industrial systems
Xcel Energy listed study provider, Minnesota compressed air and process efficiency programs

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